Jinjin Zhao

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Education

PhD **University of Chicago**, Computer Science — Sept. 2019 to June 2026

- Advisor: Sanjay Krishnan (ChiData Database Group)
- M.S. degree completed as part of Ph.D. program

BSE **Princeton University**, Computer Science — Sept. 2015 to May 2019

- Summa Cum Laude
- Minor in Statistics and Machine Learning

Experience

Google, Software Engineer — Kirkland, WA, June 2026 to Now

- Developing infrastructure and metrics for large scale Cloud TPU (Tensor Processing Unit) clusters to serve AI workloads.

Linea Labs, Research Intern — San Francisco, CA, June 2023 to Sept. 2023

- Co-designed initial MVP for Airflow reproducibility with a 6-person startup team.
- Implemented novel lineage-tracking and LLM-based code-analysis features.

Princeton Plasma Physics Lab, Research Intern — Plainsboro, NJ, June 2018 to July 2018

- Compiled data on two years of DIII-D tokamak experiments and trained neural network models to predict pedestal features driving fusion output.

Meta, Software Engineer Intern — Seattle, WA, June 2017 to Aug. 2017

- Developed a full-stack Hack (PHP) and MySQL tool for storing and retrieving test artifacts (e.g., logs and build files) from internal pre-commit automated runs.
- Scaled to over 200 million files per week, supporting a significant share of internal code development.

Meta, Facebook University Intern — Menlo Park, CA, June 2016 to Aug. 2016

- Designed and built an independent Android app with an internal music player that generated Spotify playlists based on nearby concerts.

Selected Projects

TableVault: Contextual End-to-End Management of LLM Data Pipelines (Website**) (**On-going**)**

- A open-sourced failure-safe data store for distributed AI notebooks and Python processes backed by ArangoDB.
- Enables coordinator-less control where workers can communicate execution requests (e.g. pause/resume/stop) via shared state.
- Supports rich contextual search where users can issue joint queries over AI artifacts (e.g. documents, embeddings, dataframes), and pipeline context (e.g. code, prompts, comments).

DSLog: A Compressed Query and Storage Framework for Fine-Grained Array Lineage

- Framework for fast upstream and downstream queries of cell-level lineage over machine learning operations (i.e. find the cells in the original array that contributed to cells in the transformed array).
- Augmented the Numpy library with low-level memory management (C) to record cell-to-cell operational lineage efficiently, resulting in up to 3000x improvements over Python baselines on arrays with up to 100 million cells.
- Designed a new range-based compression algorithm that improves storage space and query time of the resulting lineage graph by up to 2000x and 20x respectively.

Tracing Variation in Data Science Workflows with Jupyter Notebook Logging

- First study to quantitatively assess common conceptions in data science (e.g. data cleaning is 80% of the work).
- Designed a tool to log Jupyter notebook execution traces from 97 novices and industry experts in data science.
- Compared string search, code analysis (AST) and notebook replay techniques to analyse JSON traces for capturing user trends (e.g. most errors are resolved within 1-2 code executions).

Selected Publications

Fast Capture of Cell-Level Provenance in Numpy. 2025
Jinjin Zhao, Sanjay Krishnan
ProvenanceWeek@SIGMOD Paper

TableVault: Managing Dynamic Data Collections for LLM-Augmented Workflows 2025
Jinjin Zhao, Sanjay Krishnan
NOVAS@SIGMOD Paper

Quantifying Variation in Data Science with Fine-Grained Procedural Logging. 2024
Jinjin Zhao, Avigor Gal, Sanjay Krishnan
Under Submission Paper

Compression and In-Situ Query Processing for Fine-Grained Array Lineage. 2024
Jinjin Zhao, Sanjay Krishnan
ICDE Paper

Data Makes Better Data Scientists. 2023
Jinjin Zhao, Avidgor Gal, Sanjay Krishnan
HILDA@SIGMOD Paper

AMIR: Active Multimodal Interaction Recognition from Video and Network Traffic in Connected Environments. 2023
Shinan Liu, Tarun Mangla, Ted Shaowang, **Jinjin Zhao**, Sanjay Krishnan, Nick Feamster
UbiComp/IMWUT Paper

Data Station: Delegated, Trustworthy, and Auditable Computation to Enable Data Sharing Consortia with a Data Escrow. 2022
Siyuan Xia, Zhiru Zhu, Chris Zhu, **Jinjin Zhao**, Kyle Chard, Aaron J. Elmore, Ian Foster, Michael Franklin, Sanjay Krishnan, Raul Castro Fernandez
VLDB Paper

Towards Causal Query Answering for Debugging Video Analytics Systems. 2022
Ted Shaowang*, **Jinjin Zhao***, Stavros Sintos, Sanjay Krishnan
HILDA@SIGMOD Paper

Prediction of DIII-D Pedestal Structure From Externally Controllable Parameters. 2021
Emi Zeger, Florian Laggner, Alessandro Bortolon, Cristina Rea, Orso Meneghini, Samuli Saarelma, Brian Sammuli, Sterling Smith, **Jinjin Zhao**
IEEE Transactions on Plasma Science Paper

Activities And Awards

- **2018 - 2024 Teaching Assistant**: COS 397/497 Fall 2018 *(Princeton University)*, CMSC 16100 Autumn 2019 *(University of Chicago)*, CMSC 21800 Autumn 2020/2023 *(University of Chicago)*, DATA 13600 Spring 2024 *(University of Chicago)*
- 2020 - 2022 Curriculum and Social Minister, UChicago CS Graduate Student Ministry
- ICDE'2024 Travel Award, NSF
- 2022 University Unrestricted Fellowship, University of Chicago
- 2020 Lab Coordinator, CDAC (high school data science research summer program)
- OSDI'2020 Diversity Grant, USENIX Association
- 2019 Neubauer Graduate Scholarship, University of Chicago